

Markets take the stairs up, but the elevator down

GARCH MODELS IN R



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How?

Change the argument `distribution.model` of `ugarchspec()` from `"norm"` to `"sstd"`:

```
garchspec <- ugarchspec( mean.model = list(armaOrder = c(0, 0)),  
                          variance.model = list(model = "sGARCH"),  
                          distribution.model = "norm")
```



```
garchspec <- ugarchspec( mean.model = list(armaOrder = c(0, 0)),  
                          variance.model = list(model = "sGARCH"),  
                          distribution.model = "sstd")
```

The normal GARCH model

Under the model assumptions

$$R_t = \mu_t + e_t$$

$$e_t \sim N(0, \sigma_t^2)$$

it follows that

$$\frac{R_t - \mu_t}{\sigma_t} \sim N(0, 1)$$

Let's test

- Caveat: The normality of the standardized returns follows from an assumption
- Let's compute the standardized returns and test whether the assumption is correct.

Estimated standardized returns

- Formula

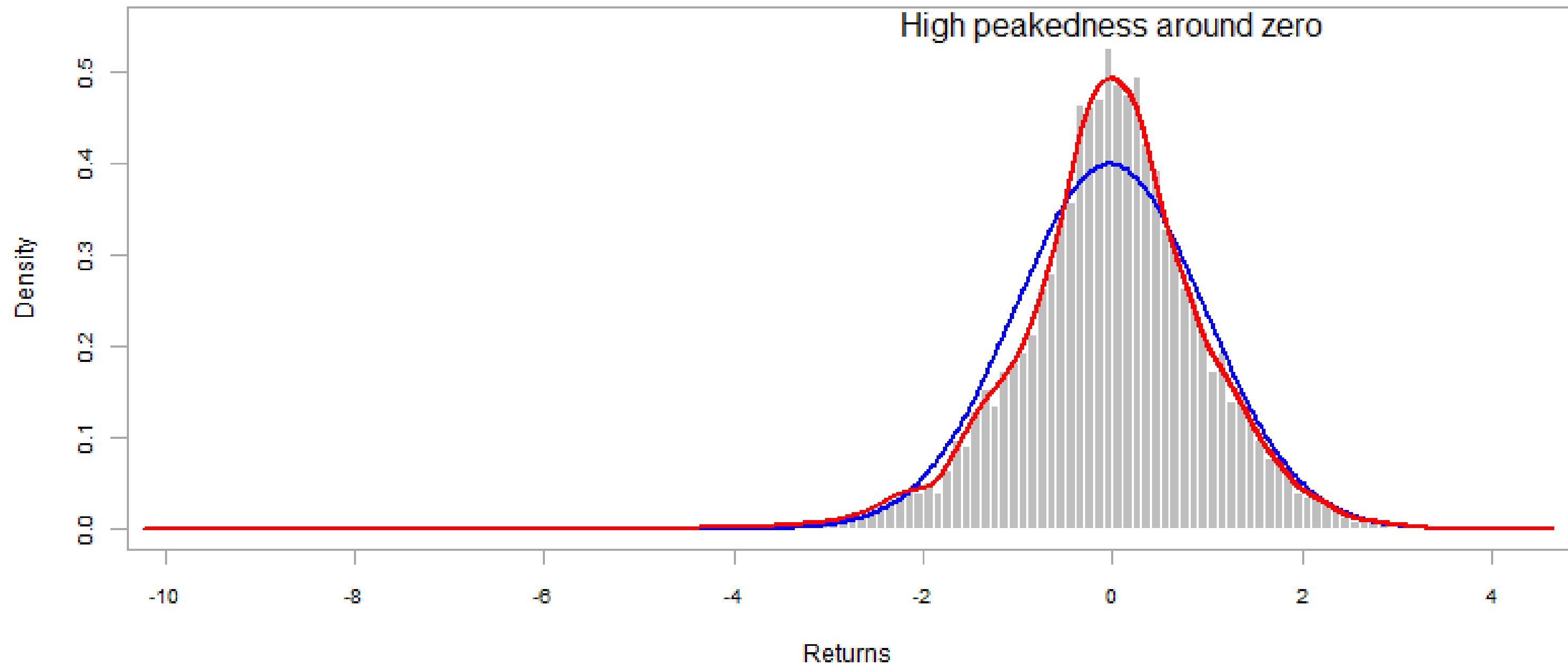
$$Z_t = \frac{R_t - \hat{\mu}_t}{\hat{\sigma}_t}$$

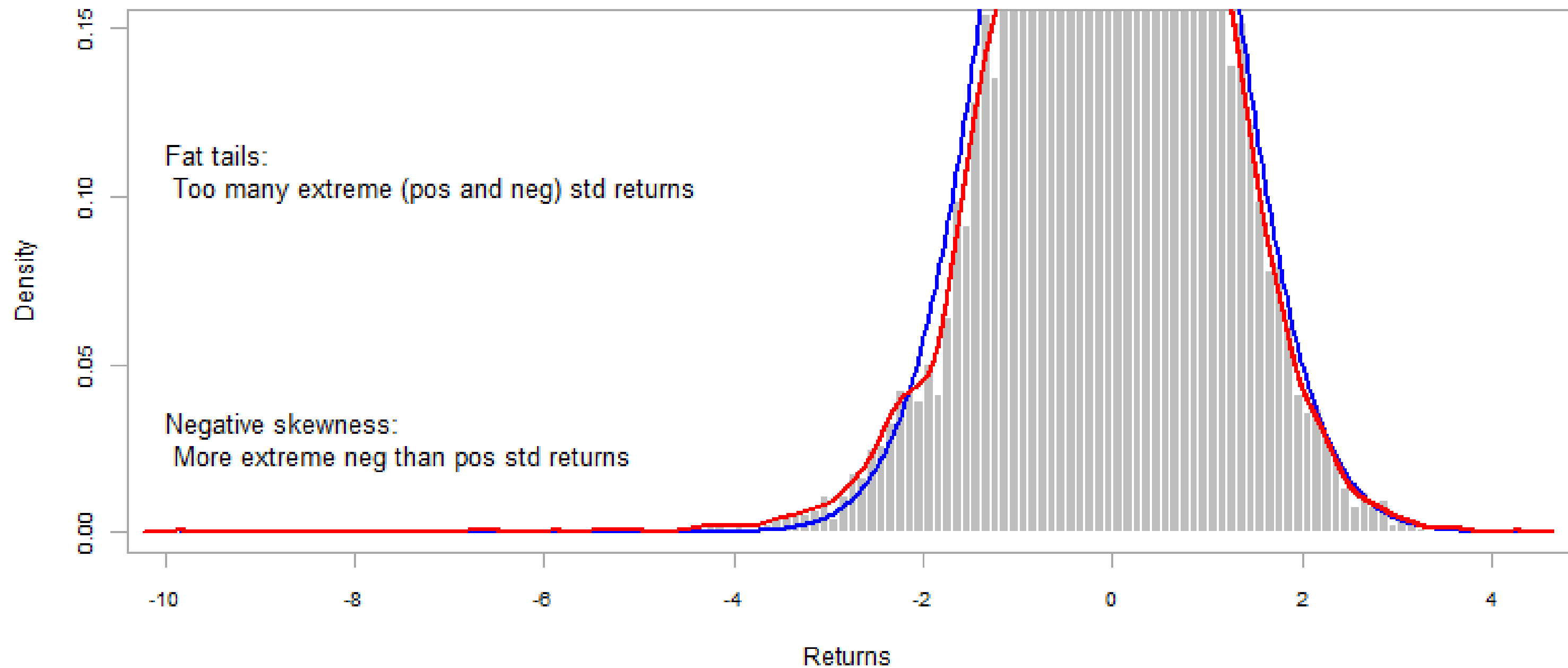
Calculation in R

```
# Obtain standardized returns  
stdret <- residuals(garchfit, standardize = TRUE)
```

Testing the normality assumption

```
library(PerformanceAnalytics)
chart.Histogram(sp500ret, methods = c("add.normal", "add.density"),
               colorset = c("gray", "red", "blue"))
```





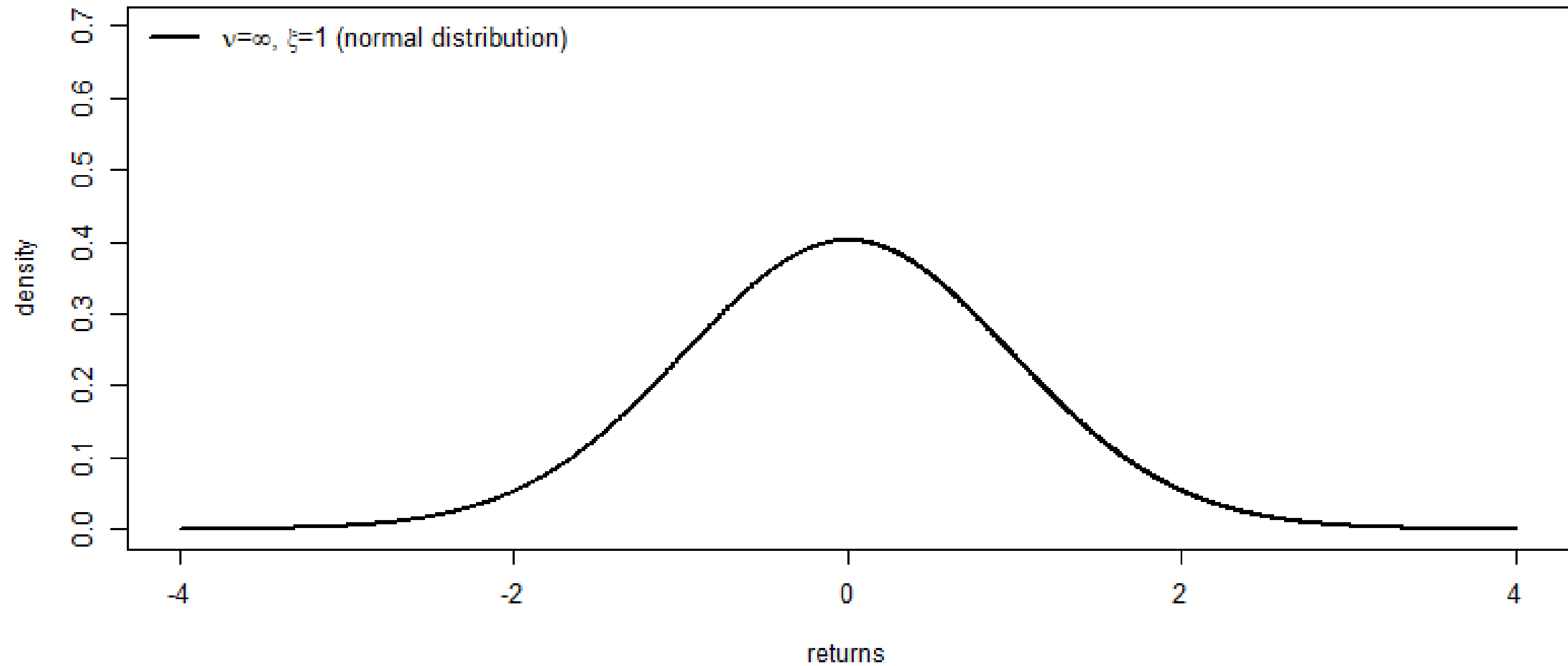
Solution

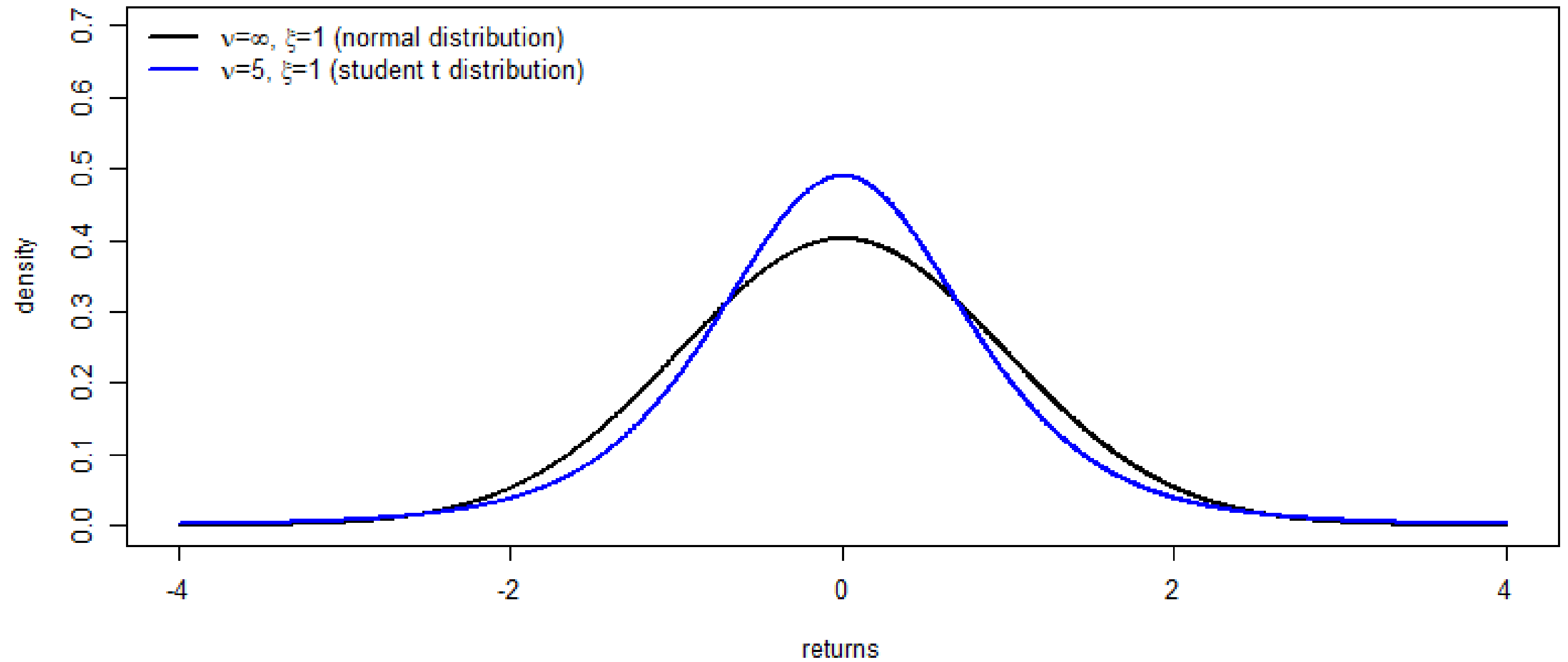
- A realistic distribution thus needs to accommodate the presence of
 - fat tails: higher probability to observe large (positive or negative) returns than under the normal distribution
 - skewness: asymmetry of the return distribution
- In `rugarch` this is possible with the **skewed student t** distribution:

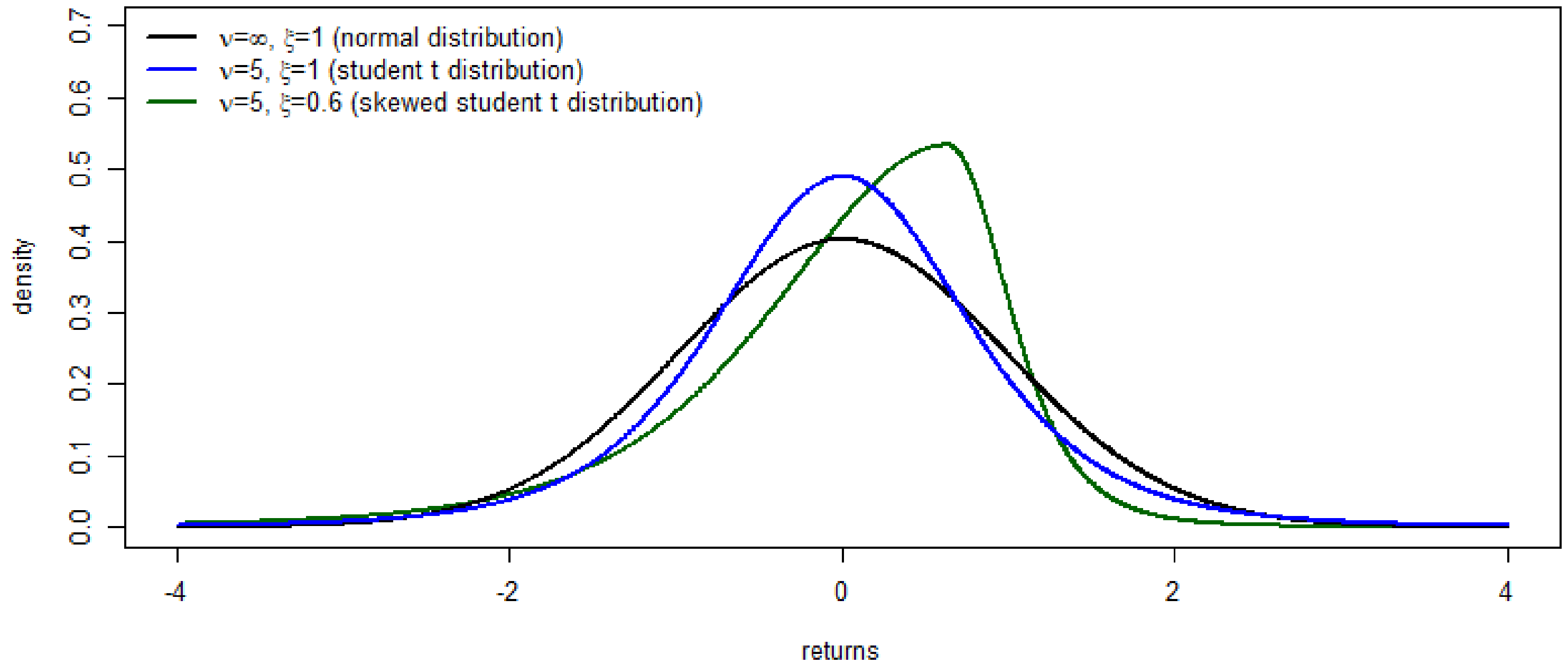
```
garchspec <- ugarchspec(distribution.model = "sstd")
```

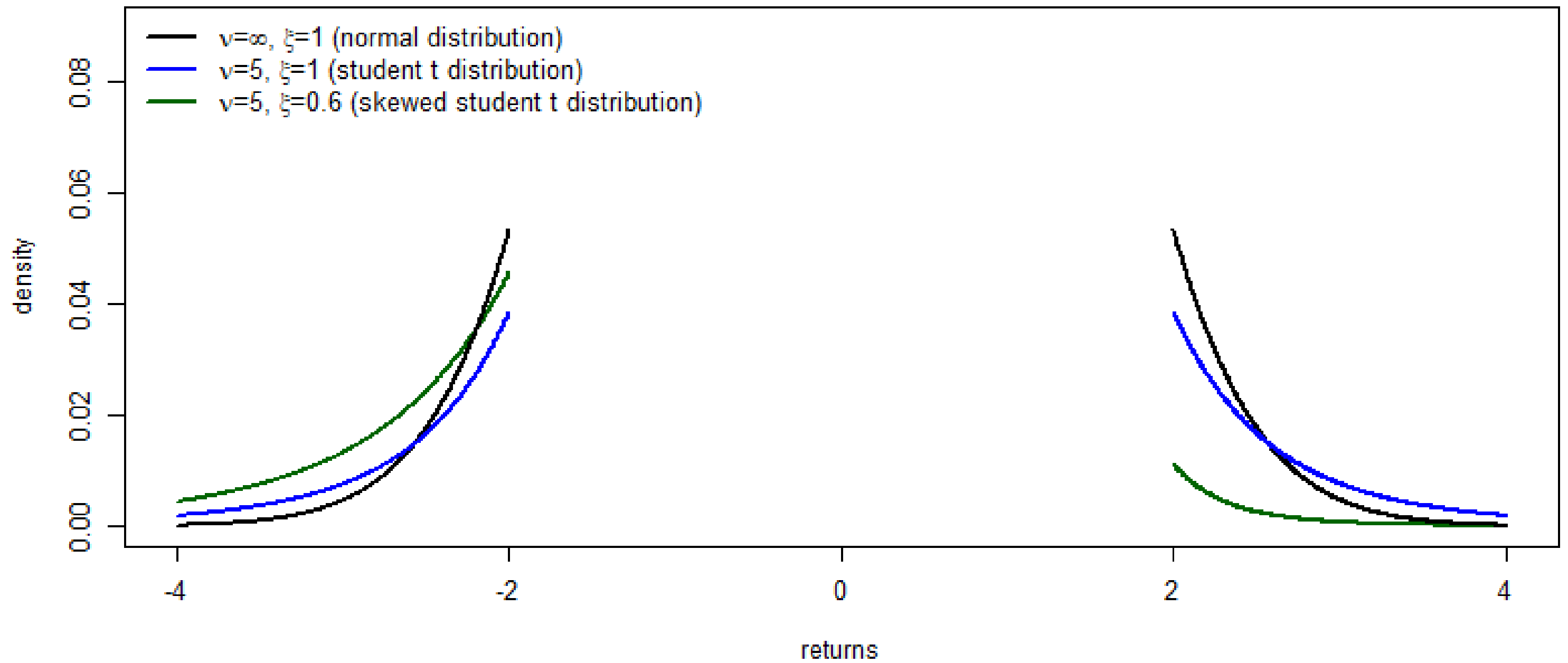
Parameters of the skewed student t distribution

- Compared to the normal distribution, the skewed student t distribution has two extra parameters:
 - Degrees of freedom parameter ν (in rugarch: `shape`): the lower is ν the fatter the tails.
 - Skewness parameter ξ (in rugarch: `skew`): when $\xi = 1$: symmetry. When $\xi < 1$: negative skewness. For $\xi > 1$: positive skewness.
- Special cases:
 - When $\nu = \infty$ and $\xi = 1$: normal distribution.
 - When $\xi = 1$: student t distribution.









GARCH model estimation with skewed student t

Set argument `distribution.model` to "sstd"

```
garchspec <- ugarchspec(mean.model = list(armaOrder = c(0,0)),  
                        variance.model = list(model = "sGARCH"),  
                        distribution.model = "sstd")
```

Estimate the model

```
garchfit <- ugarchfit(data = sp500ret, spec = garchspec)  
coef(garchfit)
```

mu	omega	alpha1	beta1	skew	shape
5.669200e-04	6.281258e-07	7.462984e-02	9.223701e-01	9.436331e-01	6.318621e+00

Let's practice!

GARCH MODELS IN R

Leverage effect

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Negative returns induce higher leverage

- $R_t < 0$
- ↓ market value
- ↑ leverage = debt / market value
- ↑ volatility

Two equations

Use separate equations for the variance following negative and positive unexpected return

$$e_t = R_t - \mu_t :$$

Case when $e_{t-1} > 0$

$$\sigma_t^2 = ???$$

Case when $e_{t-1} \leq 0$

$$\sigma_t^2 = ???$$

In case of a positive surprise

... we take the usual GARCH(1,1) equation:

Case when $e_{t-1} > 0$

$$\sigma_t^2 = \omega + \alpha e_{t-1}^2 + \beta \sigma_{t-1}^2$$

Case when $e_{t-1} \leq 0$

$$\sigma_t^2 = ???$$

In case of a negative surprise

- The predicted variance should be higher than after a positive surprise.
- This means a higher coefficient multiplying the squared prediction error, namely $\alpha + \gamma$ instead of α with $\gamma \geq 0$

Case when $e_{t-1} > 0$

$$\sigma_t^2 = \omega + \alpha e_{t-1}^2 + \beta \sigma_{t-1}^2$$

Case when $e_{t-1} \leq 0$

$$\sigma_t^2 = \omega + (\alpha + \gamma) e_{t-1}^2 + \beta \sigma_{t-1}^2$$

This is the **GJR model** proposed Glosten, Jagannathan and Runkle.

How?

Change the argument `variance.model` of `ugarchspec()` from `model = "sGARCH"` to `model = "gjrgARCH"`:

```
garchspec <- ugarchspec(mean.model = list(armaOrder = c(0, 0)),  
                        variance.model = list(model = "sGARCH"),  
                        distribution.model = "sstd")
```



```
garchspec <- ugarchspec(mean.model = list(armaOrder = c(0, 0)),  
                        variance.model = list(model = "gjrgARCH"),  
                        distribution.model = "sstd")
```

Illustration on MSFT returns

Estimate the model

```
garchfit <- ugarchfit(data = msftret, spec = garchspec)
```

Inspect the GARCH coefficients

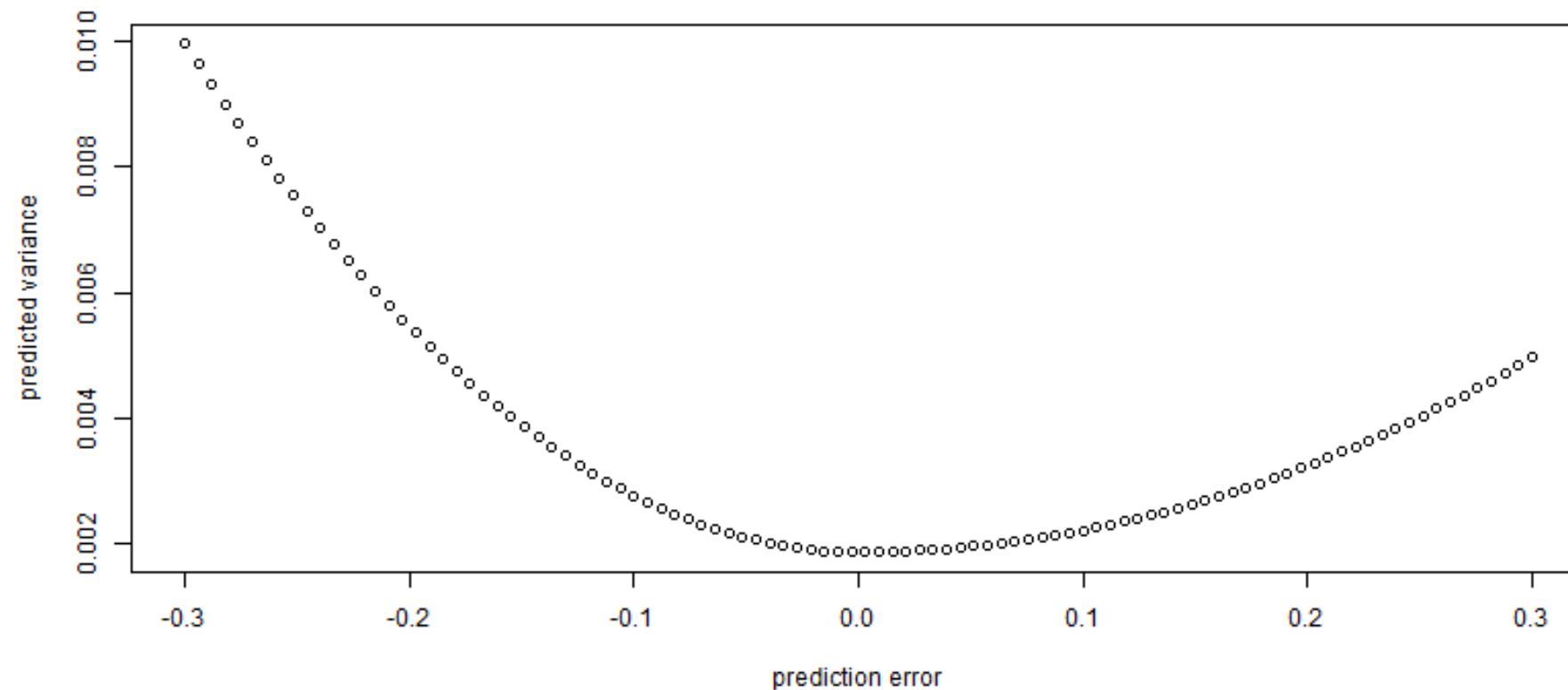
```
coef(garchfit)[2:5]
```

omega	alpha1	beta1	gamma1
2.007875e-06	3.423336e-02	9.363302e-01	5.531854e-02

$$\hat{\sigma}_t^2 = \begin{cases} 2.0 * 10^{-6} + 0.09e_{t-1}^2 + 0.94\sigma_{t-1}^2 & : e_{t-1} \leq 0 \\ 2.0 * 10^{-6} + 0.03e_{t-1}^2 + 0.94\sigma_{t-1}^2 & : e_{t-1} > 0 \end{cases}$$

Visualize volatility response using newsimpact()

```
out <- newsimpact(garchfit)
plot(out$zx, out$zy, xlab = "prediction error", ylab = "predicted variance")
```



Let's estimate a GJR GARCH model.

GARCH MODELS IN R

No pain, no gain

GARCH MODELS IN R



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GARCH-in-mean model

- Quantify the risk-reward trade-off.
- Risk: σ_t^2 . Reward: μ_t .
- GARCH-in-mean model:

$$\mu_t = \mu + \lambda \sigma_t^2$$

$\lambda > 0$ is the risk/reward parameter indicating the increase in expected return per unit of variance risk.

How?

Change the argument `mean.model` in `ugarchspec()` from `list(armaOrder = c(0, 0))` to `list(armaOrder = c(0, 0), archm = TRUE, archpow = 2)`:

```
garchspec <- ugarchspec(  
  mean.model = list(armaOrder = c(0, 0)),  
  variance.model = list(model = "gjrgARCH"),  
  distribution.model = "sstd")
```

```
garchspec <- ugarchspec(  
  mean.model = list(armaOrder = c(0, 0), archm = TRUE, archpow = 2),  
  variance.model = list(model = "gjrgARCH"),  
  distribution.model = "sstd")
```

Application to daily S&P 500 returns

Estimation

```
garchfit <- ugarchfit(data = sp500ret, spec = garchspec)
```

Inspection of estimated coefficients for the mean

```
round(coef(garchfit)[1:2], 4)
```

```
mu archm  
0.0002 1.9950
```

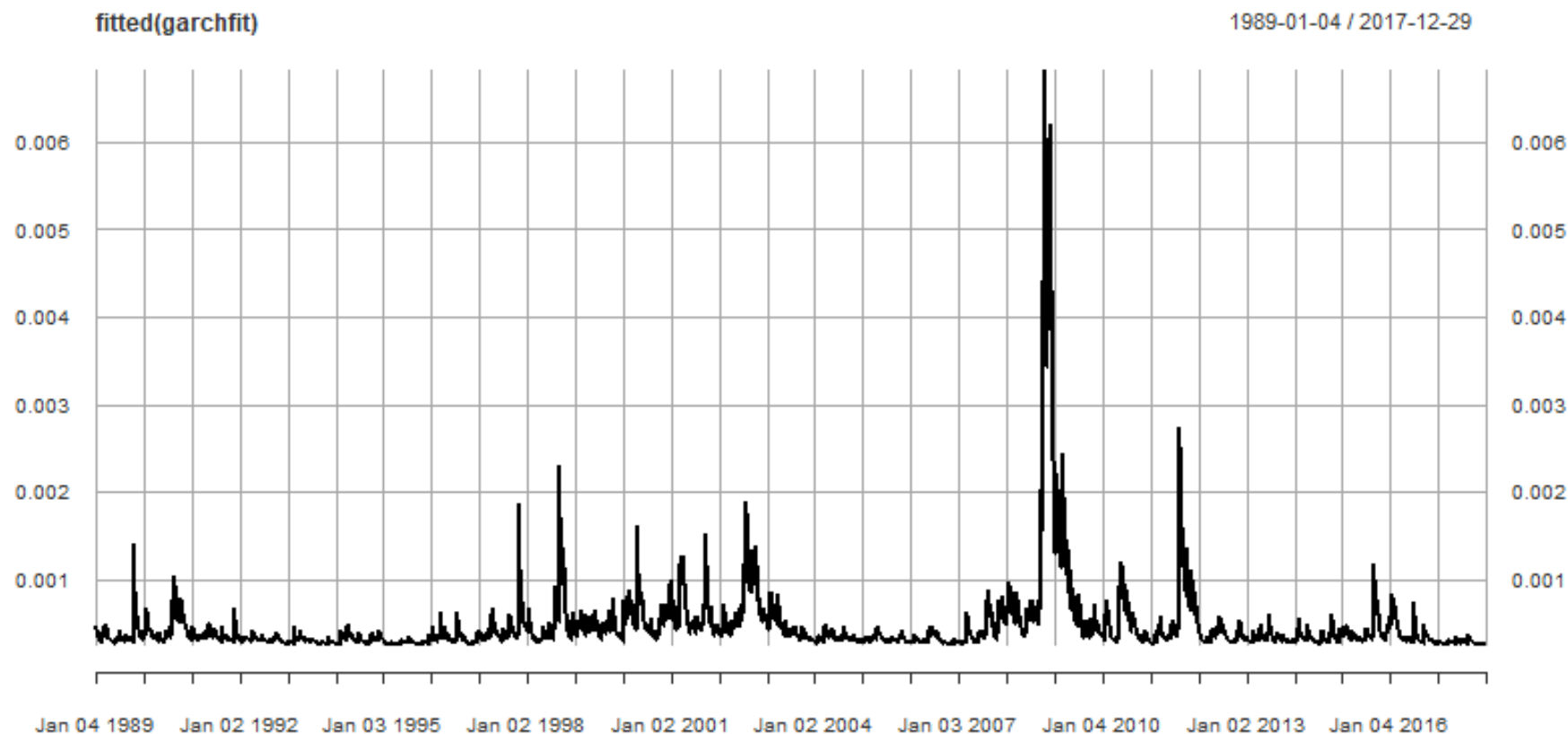
Predicted mean returns

$$\hat{\mu}_t = 0.0002 + 1.9950\hat{\sigma}_t^2$$

Time series plot of predicted returns

- Plot them in R

```
plot(fitted(garchfit))
```



Today's return predicts tomorrow's return

- The GARCH-in-mean uses the financial theory of a risk-reward trade-off to build a conditional mean model.
- Let's now use statistical theory to make a mean model that exploits the correlation between today's return and tomorrow's return.
- The most popular model is the AR(1) model:
 - AR(1) stands for autoregressive model of order 1.
 - It predicts the next return using the deviation of the return from its long term mean value μ :

$$\mu_t = \mu + \rho(R_{t-1} - \mu)$$

A positive autoregressive coefficient

$$\mu_t = \mu + \rho(R_{t-1} - \mu)$$

- $\rho > 0$:
 - A higher (resp. lower) than average return is followed by a higher (resp. lower) than average return.
 - Possible explanation: markets underreact to news and hence there is momentum in returns.
- $|\rho| < 1$: Mean reversion: The deviations of R_t from μ are transitory.

A negative autoregressive coefficient

$$\mu_t = \mu + \rho(R_{t-1} - \mu)$$

- $\rho < 0$:
 - A higher (resp. lower) than average return is followed by a lower (resp. higher) than average return.
 - Possible explanation: markets overreact to news and hence there is reversal in returns.

Application to daily S&P 500 returns

Specification and estimation of AR(1)-GJR GARCH with sst distribution

```
garchspec <- ugarchspec(  
  mean.model = list(armaOrder = c(1, 0)),  
  variance.model = list(model = "gjrGARCH"),  
  distribution.model = "sstd")  
garchfit <- ugarchfit(data = sp500ret, spec = garchspec)
```

Estimates of the AR(1) model

```
round(coef(garchfit)[1:2], 4)
```

mu	ar1
0.0003	-0.0292

MA(1) and ARMA(1,1) model

The Moving Average model of order 1 uses the deviation of the return from its conditional mean:

$$\mu_t = \mu + \theta(R_{t-1} - \mu_{t-1})$$

ARMA(1,1) combines AR(1) and MA(1):

$$\mu_t = \mu + \rho(R_{t-1} - \mu) + \theta(R_{t-1} - \mu_{t-1})$$

How?

MA(1)

```
garchspec <- ugarchspec(  
  mean.model = list(armaOrder = c(0, 1)),  
  variance.model = list(model = "gjrgARCH"),  
  distribution.model = "sstd")
```

ARMA(1, 1)

```
garchspec <- ugarchspec(  
  mean.model = list(armaOrder = c(1, 1)),  
  variance.model = list(model = "gjrgARCH"),  
  distribution.model = "sstd")
```

Your turn to change the `mean.model` argument

GARCH MODELS IN R

Avoid unnecessary complexity

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Avoid unneeded complexity

- If you know
 - The mean dynamics are negligible
 - There is no leverage effect in the variance
 - The distribution is symmetric and fat-tailed

Then a constant mean, standard GARCH(1, 1) with student t distribution is an appropriate specification to use:

```
garchspec <- ugarchspec(mean.model = list(armaOrder = c(0, 0)),  
                        variance.model = list(model = "sGARCH"),  
                        distribution.model = "std")
```


Restrict the parameter estimates

- If you know that the parameters
 - are equal to a certain value
 - or, are inside an interval
- Then you should impose this in the specification using the methods
 - `setfixed()`
 - `setbounds()`

Application to exchange rates

Specification and estimation

```
garchspec <- ugarchspec(mean.model = list(armaOrder = c(0, 0)),  
                        variance.model = list(model = "sGARCH"),  
                        distribution.model = "std")  
garchfit <- ugarchfit(data = EURUSDret, spec = garchspec)
```

Estimation results

```
coef(garchfit)
```

mu	omega	alpha1	beta1	shape
-3.562136e-05	8.005123e-08	3.097322e-02	9.674496e-01	8.821902e+00

Example of setfixed()

- If you know `alpha1 = 0.05` and `shape = 6` : impose those values in the estimation.
- How? Use of `setfixed()` method on a `ugarchspec` object

```
setfixed(garchspec) <- list(alpha1 = 0.05, shape = 6)
```

Result

```
garchfit <- ugarchfit(data = EURUSDret, spec = garchspec)
```

```
coef(garchfit)
```

mu	omega	alpha1	beta1	shape
-4.142922e-05	2.061772e-07	5.000000e-02	9.489622e-01	6.000000e+00

Bounds on parameters

- The GARCH parameters can be restricted to an interval.
- Sometimes the interval of plausible values is large:
 - To ensure the variance is positive, we require e.g. that all variance parameters (ω , α , β , γ) are positive.
- Sometimes the interval of plausible values is smaller:
 - Likely values of α are in between 0.05 and 0.2
 - Likely values of β are in between 0.7 and 0.95
- Such bound constraints on the parameters can be imposed using the `setbounds()` method.

Example of setbounds()

```
setbounds(garchspec) <- list(alpha1 = c(0.05, 0.2), beta1 = c(0.8, 0.95))
```

Use your intuition to avoid unneeded complexity.

Use the information you have:

- to build simple (and smart) models
- to fix parameter values or set bounds
- to make the GARCH dynamics realistic:
 - mean reversion of the volatility around the sample standard deviation

```
sd(EURUSDret) # returns a value of 0.006194049
```

Volatility clusters and mean reversion of volatility



Variance targeting

- Mathematically, this means that the unconditional variance implied by the GARCH models equals the sample variance $\hat{\sigma}^2$.
- How? By setting the argument `variance.targeting = TRUE` in `variance.model` of `ugarchspec()` :

```
garchspec <- ugarchspec(mean.model = list(armaOrder = c(0,0)),  
                        variance.model = list(model = "sGARCH",  
                                              variance.targeting = TRUE),  
                        distribution.model = "std")  
  
garchfit <- ugarchfit(data = EURUSDret, spec = garchspec)  
all.equal(uncvariance(garchfit), sd(EURUSDret) ^ 2, tol = 1e-4)
```

TRUE

Let's impose restrictions on the GARCH model

GARCH MODELS IN R